RANDOM VARIABLES

Lect. İsmail BAŞOĞLU, Ph.D.

Özyeğin University

February 2, 2024

1. The Random Variable Concept
   - What Is A Random Variable?
   - Discrete RVs vs. Continuous RVs

1. The Random Variable Concept
   - What Is A Random Variable?
   - Discrete RVs vs. Continuous RVs
2. Discrete Random Variables (DRVs)
   - Probability Mass Function (PMF)
   - Cumulative Distribution Function (CDF)
   - Expected Value of A Discrete Random Variable
   - Variance of A Discrete Random Variable

1. The Random Variable Concept
   - What Is A Random Variable?
   - Discrete RVs vs. Continuous RVs
2. Discrete Random Variables (DRVs)
   - Probability Mass Function (PMF)
   - Cumulative Distribution Function (CDF)
   - Expected Value of A Discrete Random Variable
   - Variance of A Discrete Random Variable
3. Continuous Random Variables (CRVs)
   - Probability Density Function (PDF)
   - Cumulative Distribution Function (CDF)
   - Expected Value of A Continuous Random Variable
   - Variance of A Continuous Random Variable

1. The Random Variable Concept
   - What Is A Random Variable?
   - Discrete RVs vs. Continuous RVs
2. Discrete Random Variables (DRVs)
   - Probability Mass Function (PMF)
   - Cumulative Distribution Function (CDF)
   - Expected Value of A Discrete Random Variable
   - Variance of A Discrete Random Variable
3. Continuous Random Variables (CRVs)
   - Probability Density Function (PDF)
   - Cumulative Distribution Function (CDF)
   - Expected Value of A Continuous Random Variable
   - Variance of A Continuous Random Variable
4. Important Identities of Expectation and Variance
   - Expected Value of A Linear Transformation
   - Variance of A Linear Transformation

What is A Random Variable?

- Consider an experiment with sample space S.
  - Frequently, we are interested in some **quantitative function** of the outcome as opposed to the actual outcome itself.
- **Random Variable (RV):** A real-valued function defined on the sample space.
- Consider the following example.
  - **Experiment:** Tossing two dice
  - **Sample Space:** All 36 equally likely outcomes.
  - **Random Variable X:** Sum of two dice.
  - **Random Variable Y:** Absolute difference between two dice.
  - **Random Variable T:** Maximum of the two dice.
  - **Random Variable W:** Minimum of the two dice.

Probabilities on Random Variables

- Because the value of a random variable is determined by the outcome of the experiment, we may assign probabilities to the possible values of the random variable.
- Letting X denote the random variable that is defined as the sum of two fair dice; then

$$
\begin{aligned}
&P\{X=2\}=P\{(1,1)\}=\tfrac{1}{36},\\
&P\{X=3\}=P\{(1,2),(2,1)\}=\tfrac{2}{36},\\
&P\{X=4\}=P\{(1,3),(2,2),(3,1)\}=\tfrac{3}{36},\\
&P\{X=5\}=P\{(1,4),(2,3),(3,2),(4,1)\}=\tfrac{4}{36},\\
&P\{X=6\}=P\{(1,5),(2,4),(3,3),(4,2),(5,1)\}=\tfrac{5}{36},\\
&P\{X=7\}=P\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}=\tfrac{6}{36},\\
&P\{X=8\}=P\{(2,6),(3,5),(4,4),(5,3),(6,2)\}=\tfrac{5}{36},\\
&P\{X=9\}=P\{(3,6),(4,5),(5,4),(6,3)\}=\tfrac{4}{36},\\
&P\{X=10\}=P\{(4,6),(5,5),(6,4)\}=\tfrac{3}{36},\\
&P\{X=11\}=P\{(5,6),(6,5)\}=\tfrac{2}{36},\\
&P\{X=12\}=P\{(6,6)\}=\tfrac{1}{36}.
\end{aligned}
$$

Probabilities on Random Variables

- **Probabilities on Random Variables:**
  - S is the sample space, set of all possible outcomes.
  - X, random variable, is a function that relates the outcomes with real numbers.
  - $P : \mathbb{R} \rightarrow [0, 1]$, probability is a function from real numbers to unit interval.

Simple Random Variable Example

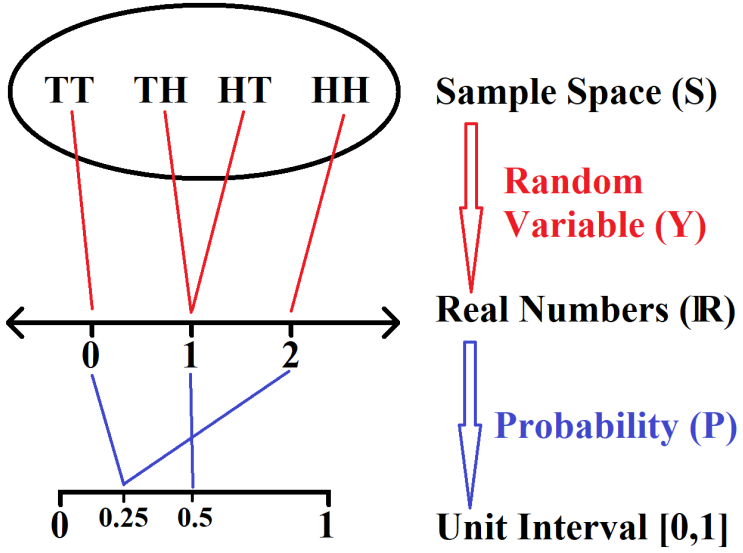
Q1. Suppose we toss two fair coins. Letting Y denote the number of heads appearing, then what values can Y take with what probabilities?

Simple Random Variable Example

Q1. Suppose we toss two fair coins. Letting Y denote the number of heads appearing, then what values can Y take with what probabilities?

A1. Y can take values 0, 1, and 2.

$$
\begin{aligned}
P\{Y=0\} &= P\{TT\} \\
&= 0.25 \\
P\{Y=1\} &= P\{TH, HT\} \\
&= 0.5 \\
P\{Y=2\} &= P\{HH\} \\
&= 0.25
\end{aligned}
$$



Simple Random Variable Example II

Q2. Suppose that our experiment consists of tossing 3 fair coins. If we let Y denote the number of heads that appear, what are the probabilities of random variable Y on the values that it can take?

Simple Random Variable Example II

Q2. Suppose that our experiment consists of tossing 3 fair coins. If we let Y denote the number of heads that appear, what are the probabilities of random variable Y on the values that it can take?

A2. Y can take values 0, 1, 2, and 3.

$$P\{Y=3\} = P\{(HHH)\} = \tfrac{1}{8} = 0.125$$
$$P\{Y=2\} = P\{(THH),(HTH),(HHT)\} = \tfrac{3}{8} = 0.375$$
$$P\{Y=1\} = P\{(TTH),(THT),(HTT)\} = \tfrac{3}{8} = 0.375$$
$$P\{Y=0\} = P\{(TTT)\} = \tfrac{1}{8} = 0.125$$

Note that, Y can only take one of these values, thus:

$$P\{Y=0\} + P\{Y=1\} + P\{Y=2\} + P\{Y=3\} = 1$$

Simple Random Variable Example III

Q3. In an urn, there are 20 balls, 3 of which are golden. We pay $\$1$ and randomly choose three balls from the urn. We win respectively $\$1$, $\$50$, or $\$1000$ if we select a single golden ball, two, or three golden balls. If X denotes the profit from this game, what are the probabilities of X on the values that it can take?

Simple Random Variable Example III

Q3. In an urn, there are 20 balls, 3 of which are golden. We pay $\$1$ and randomly choose three balls from the urn. We win respectively $\$1$, $\$50$, or $\$1000$ if we select a single golden ball, two, or three golden balls. If X denotes the profit from this game, what are the probabilities of X on the values that it can take?

A3. X can take values -1, 0, 49, and 999.

$$P\{X=-1\}=\frac{\binom{3}{0}\binom{17}{3}}{\binom{20}{3}}=\frac{680}{1140},\quad P\{X=0\}=\frac{\binom{3}{1}\binom{17}{2}}{\binom{20}{3}}=\frac{408}{1140},$$
$$P\{X=49\}=\frac{\binom{3}{2}\binom{17}{1}}{\binom{20}{3}}=\frac{51}{1140},\quad P\{X=999\}=\frac{\binom{3}{3}\binom{17}{0}}{\binom{20}{3}}=\frac{1}{1140}$$

Discrete RVs vs. Continuous RVs

- Mainly, random variables are studied under two classes:
  - Discrete random variables.
  - Continuous random variables.
- Discrete and continuous RVs are conceptualized and, thus, modeled differently.

Discrete Random Variables

- **Discrete Random Variables (DRVs):** A RV is a discrete RV if it takes a countable[1] number of possible values.
- **Examples:**
  - X: Number of heads in three consecutive coin tosses.
  $$X \in \{0,1,2,3\}$$
  - Y: Number of clients entering a bank between 8 AM and 5 PM.
  $$Y \in \{0,1,2,3,4,\ldots\}$$
  - T: League points gained after a football match.
  $$T \in \{0,1,3\}$$

[1] Check "LS01.pdf" for the definition.

Describing Discrete Random Variables

- In order to describe a **discrete** random variable X mathematically, we use the following functions that belong to X:
  i. Probability **mass** function of X.
  ii. Cumulative distribution function of X.
  iii. Expected value of X.
  iv. Variance of X.

Probability Mass Function (PMF)

- For a discrete random variable X, the **probability mass function** is defined as:

$$p(x) = P\{X = x\}$$

- It shows the probabilities of a discrete random variable taking on its possible values.
- Probability mass function is often abbreviated as **PMF**.

Probability Mass Function: Example

Q4. A gambler plays a game in which s/he will lose \$1 with prob. 0.3, break even with probability 0.5, and gain \$2 with probability 0.2. Let X denote the amount of money that is gained in this gamble. Write down and draw the PMF of the random variable X.

Probability Mass Function: Example

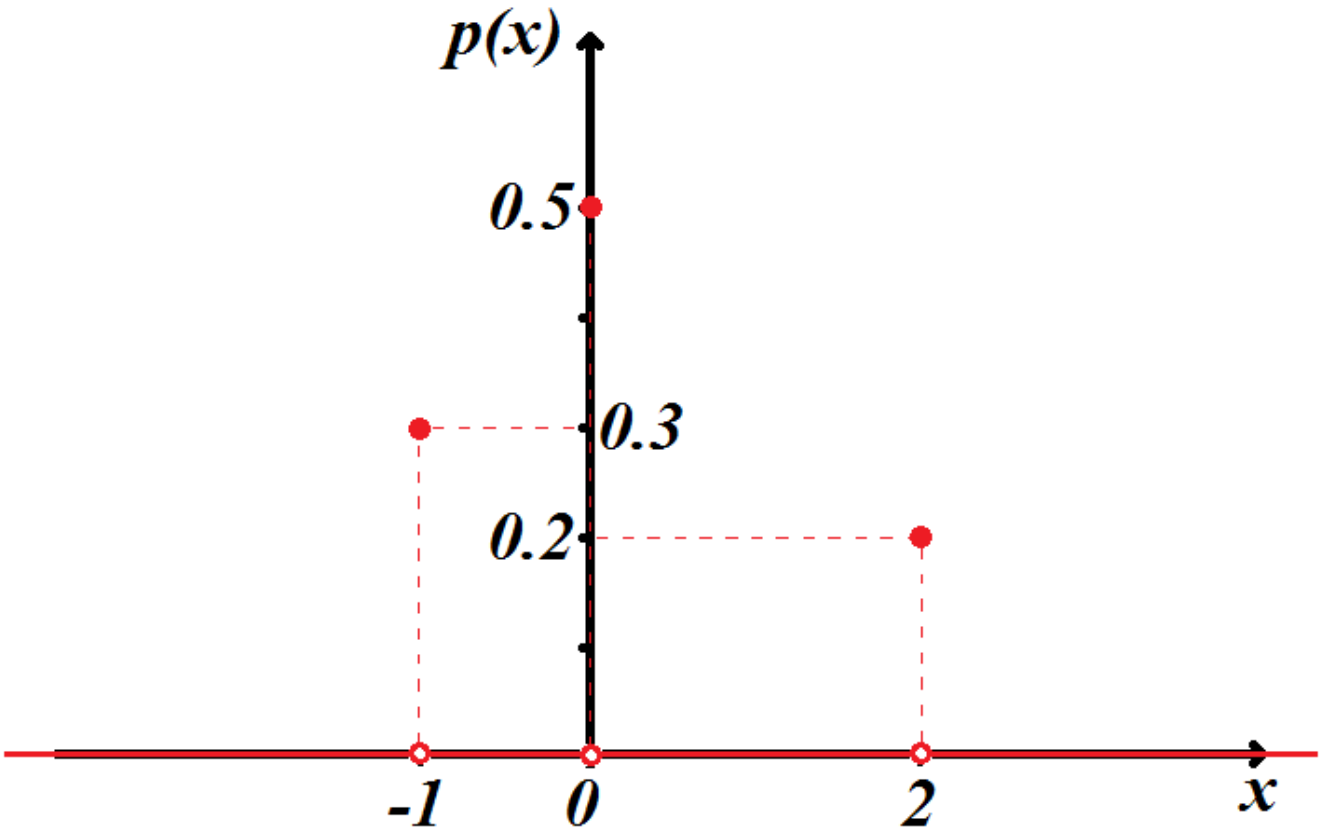
Q4. A gambler plays a game in which s/he will lose \$1 with prob. 0.3, break even with probability 0.5, and gain \$2 with probability 0.2. Let X denote the amount of money that is gained in this gamble. Write down and draw the PMF of the random variable X.

A4. The probability mass function $p(x)$ of X is:

$$p(x) = P\{X = x\} = \begin{cases} 0.3, & x = -1 \\ 0.5, & x = 0 \\ 0.2, & x = 2 \\ 0, & \text{elsewhere} \end{cases}$$

Probability Mass Function: Example (cont'd)

A4. The probability mass function can be drawn as follows:



Probability Mass Function: Example II

Q5. Write down and draw the probability mass function of the random variable representing the sum when two dice are rolled.

Probability Mass Function: Example II

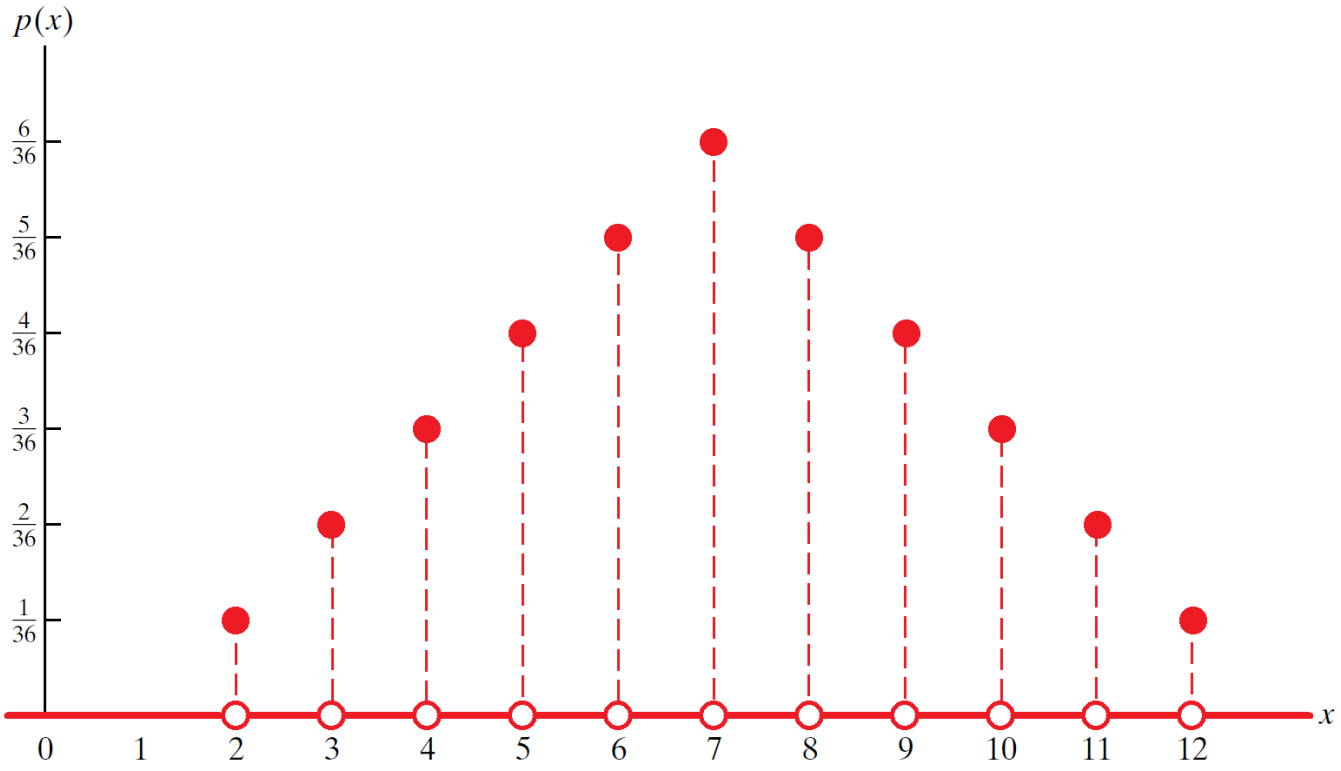
Q5. Write down and draw the probability mass function of the random variable representing the sum when two dice are rolled.

A5. We have already shown the PMF:

$$p(x) = \begin{cases} 1/36, & x = 2, 12 \\ 2/36, & x = 3, 11 \\ 3/36, & x = 4, 10 \\ 4/36, & x = 5, 9 \\ 5/36, & x = 6, 8 \\ 6/36, & x = 7 \\ 0, & \text{elsewhere} \end{cases}$$

Probability Mass Function: Example II (cont'd)

A5. The probability mass function can be drawn as follows:



Important Properties of A PMF

- A PMF is a **non-negative** function.
- For a countable number of real values, the PMF returns positive values.
  - These values sum up to 1.
- For the remaining values, the PMF is equal to zero.

Probability Mass Function: Example III

Q6. For each of the following, determine the constant c so that $p(x)$ satisfies the conditions of being a probability mass function.

a. $p(x) = cx, \quad x \in \{1,2,3,4,5,6\}$
b. $p(x) = c(x+1)^2, \quad x \in \{0,1,2\}$
c. $p(x) = c\left(x^2+1\right), \quad x \in \{-1,0,1\}$

A6. In order to determine the constants, we will use the fact that the positive probabilities in a PMF should sum up to 1.

Probability Mass Function: Example III (cont'd)

a. $p(x) = cx, \quad x \in \{1,2,3,4,5,6\}$

$$c + 2c + 3c + 4c + 5c + 6c = 21c = 1 \Rightarrow c = \frac{1}{21}$$

b. $p(x) = c(x+1)^2, \quad x \in \{0,1,2\}$

$$c(0+1)^2 + c(1+1)^2 + c(2+1)^2 = 14c = 1 \Rightarrow c = \frac{1}{14}$$

c. $p(x) = c\left(x^2+1\right), \quad x \in \{-1,0,1\}$

$$c\left((-1)^2+1\right) + c\left(0^2+1\right) + c\left(1^2+1\right) = 5c = 1 \Rightarrow c = 0.2$$

Cumulative Distribution Function (CDF)

- For a discrete random variable X, the **cumulative distribution function** is defined as:

$$F(x) = P\{X \leq x\} = \sum_{\substack{i:p(i)>0 \\ i \leq x}} p(i), \quad -\infty < x < \infty$$

- It specifies, for all real values x, the probability of the random variable X to be less than or equal to a certain real value x.
- Cumulative distribution function is often abbreviated as **CDF**.

Cumulative Distribution Function: Example

Q7. Suppose X has a PMF given by:

$$p(x) = \begin{cases} 0.25, & x = 1 \\ 0.5, & x = 2 \\ 0.125, & x = 3 \\ 0.125, & x = 4 \\ 0, & \text{elsewhere} \end{cases}$$

Write and draw the CDF of X?

Cumulative Distribution Function: Example

Q7. Suppose X has a PMF given by:

$$p(x) = \begin{cases} 0.25, & x = 1 \\ 0.5, & x = 2 \\ 0.125, & x = 3 \\ 0.125, & x = 4 \\ 0, & \text{elsewhere} \end{cases}$$

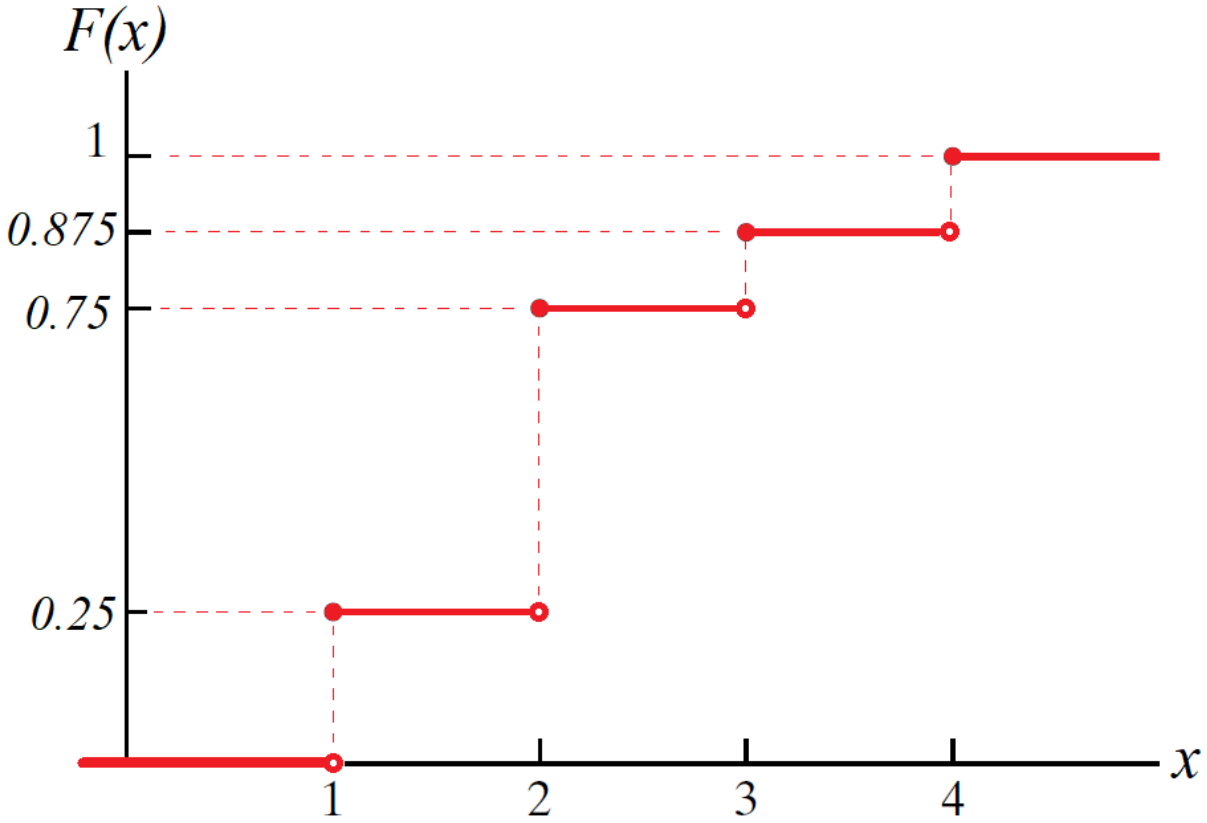
Write and draw the CDF of X?

A7. The following function is the CDF of the random variable X.

$$F(x) = \begin{cases} 0, & x < 1 \\ 0.25, & 1 \leq x < 2 \\ 0.75, & 2 \leq x < 3 \\ 0.875, & 3 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$

Cumulative Distribution Function: Example (cont'd)

A7. The cumulative distribution function can be drawn as follows:



Important Properties of A CDF

For a discrete random variable:

- A CDF is a non-decreasing function:
  - Suppose $a, b \in \mathbb{R}$ such that $a \leq b$.
  - Because the event $X \leq a$ is contained in the event $X \leq b$:

$$P\{X \leq a\} \leq P\{X \leq b\}, \Rightarrow F(a) \leq F(b)$$

- When you draw a CDF, it always starts from zero and ends up at one:
  - $\lim_{x \to -\infty} F(x) = P\{X < -\infty\} = 0$
  - $\lim_{x \to \infty} F(x) = P\{X < \infty\} = 1$
- It is a **stair-shaped** function.

Cumulative Distribution Function: Example II

Q8. We toss two fair coins and Y denotes the number of heads appearing, write down and draw the PMF and the CDF of Y?

Cumulative Distribution Function: Example II

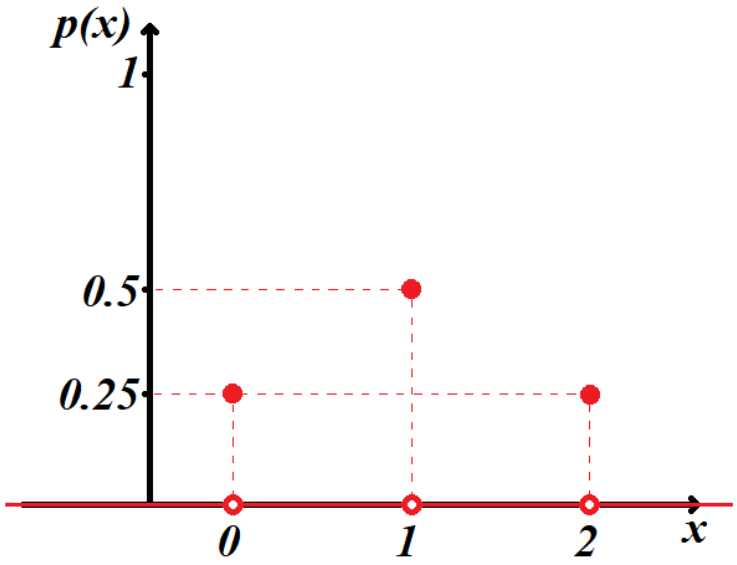
Q8. We toss two fair coins and Y denotes the number of heads appearing, write down and draw the PMF and the CDF of Y?

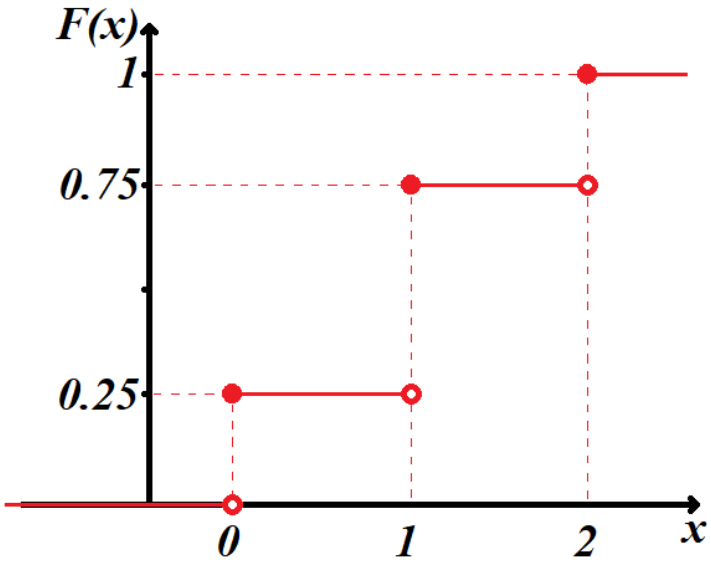
A8. The PMF and the CDF can be written as follows:

$$p(x) = \begin{cases} 0.25, & x = 0 \\ 0.5, & x = 1 \\ 0.25, & x = 2 \\ 0, & \text{elsewhere} \end{cases} \qquad F(x) = \begin{cases} 0, & x < 0 \\ 0.25, & 0 \leq x < 1 \\ 0.75, & 1 \leq x < 2 \\ 1, & \text{x} \geq 2 \end{cases}$$

Cumulative Distribution Function: Example II (cont'd)

A8. The PMF and the CDF can be drawn as follows:





Expected Value of A Discrete Random Variable

- Expected value is an important concept in probability theory.
- For a discrete random variable X with PMF $p(x)$, the **expected value (expectation)** is defined as:

$$E[X] = \sum_{x:p(x)>0} xP\{X = x\} = \sum_{x:p(x)>0} xp(x)$$

- In words, the expected value of X is:
  - a weighted average of the possible values that X can take on,
  - where the weights are the probabilities that X assumes.
- Expected value is the **average (mean) value** that the random variable X can take.

Expected Value of A Discrete Random Variable: Example

Q9. Find $E[X]$, where X is the outcome when we roll a fair die.

Expected Value of A Discrete Random Variable: Example

Q9. Find $E[X]$, where X is the outcome when we roll a fair die.

A9. At first, let us derive the PMF of X:

$$p(x) = \begin{cases} \frac{1}{6}, & x = 1,2,3,4,5,6 \\ 0, & \text{elsewhere} \end{cases}$$

Expected Value of A Discrete Random Variable: Example

Q9. Find $E[X]$, where X is the outcome when we roll a fair die.

A9. At first, let us derive the PMF of X:

$$p(x) = \begin{cases} \frac{1}{6}, & x = 1,2,3,4,5,6 \\ 0, & \text{elsewhere} \end{cases}$$

Now, we can compute the expected value of X:

$$\begin{aligned} E[X] &= \sum_{x=1}^{6} xp(x) \\ &= 1 \times \tfrac{1}{6} + 2 \times \tfrac{1}{6} + 3 \times \tfrac{1}{6} + 4 \times \tfrac{1}{6} + 5 \times \tfrac{1}{6} + 6 \times \tfrac{1}{6} \\ &= \tfrac{21}{6} = 3.5 \end{aligned}$$

Expected Value of A Discrete Random Variable: Ex. II

Q10. A school class of 120 students is driven in 3 buses to a symphonic performance. There are 36 students in one of the buses, 40 in another, and 44 in the third bus. When the buses arrive, one of the 120 students is randomly chosen. Let X denote the number of students on the bus of that randomly chosen student, and find $E[X]$.

Expected Value of A Discrete Random Variable: Ex. II

Q10. A school class of 120 students is driven in 3 buses to a symphonic performance. There are 36 students in one of the buses, 40 in another, and 44 in the third bus. When the buses arrive, one of the 120 students is randomly chosen. Let X denote the number of students on the bus of that randomly chosen student, and find $E[X]$.

A10. The random variable X can take values 36, 40, or 44, and the respective probabilities are proportional to these values as well.

$$p(36) = \frac{36}{120} = \frac{3}{10}, \quad p(40) = \frac{40}{120} = \frac{1}{3}, \quad p(44) = \frac{44}{120} = \frac{11}{30}$$

Expected Value of A Discrete RV: Ex. II (cont'd)

A10. The PMF of X is:

$$p(x) = \begin{cases} 3/10, & x = 36 \\ 1/3, & x = 40 \\ 11/30, & x = 44 \\ 0, & \text{elsewhere} \end{cases}$$

Hence, the expected value of X can be calculated by:

$$\begin{aligned} E[X] &= 36 \times \tfrac{3}{10} + 40 \times \tfrac{1}{3} + 44 \times \tfrac{11}{30} = \tfrac{108}{10} + \tfrac{40}{3} + \tfrac{484}{30} \\ &= \tfrac{324}{30} + \tfrac{400}{30} + \tfrac{484}{30} = \tfrac{1208}{30} \approx 40.2667 \end{aligned}$$

Expected Value of A Discrete Random Variable: Ex. III

Q11. Find the expectation of the following three random variables W, X, and Y:

Expected Value of A Discrete Random Variable: Ex. III

Q11. Find the expectation of the following three random variables W, X, and Y:

a. $p_W(w) = \begin{cases} 1, & w = 0 \\ 0, & \text{elsewhere} \end{cases}$

Expected Value of A Discrete Random Variable: Ex. III

Q11. Find the expectation of the following three random variables W, X, and Y:

a. $p_W(w) = \begin{cases} 1, & w = 0 \\ 0, & \text{elsewhere} \end{cases}$

b. $p_X(x) = \begin{cases} 0.5, & x = -1 \\ 0.5, & x = 1 \\ 0 & \text{elsewhere} \end{cases}$

Expected Value of A Discrete Random Variable: Ex. III

Q11. Find the expectation of the following three random variables W, X, and Y:

a. $p_W(w) = \begin{cases} 1, & w = 0 \\ 0, & \text{elsewhere} \end{cases}$

b. $p_X(x) = \begin{cases} 0.5, & x = -1 \\ 0.5, & x = 1 \\ 0 & \text{elsewhere} \end{cases}$

c. $p_Y(y) = \begin{cases} 0.5, & y = -100 \\ 0.5, & y = 100 \\ 0 & \text{elsewhere} \end{cases}$

Expected Value of A Discrete Random Variable: Ex. III

Q11. Find the expectation of the following three random variables W, X, and Y:

a. $p_W(w) = \begin{cases} 1, & w = 0 \\ 0, & \text{elsewhere} \end{cases}$

b. $p_X(x) = \begin{cases} 0.5, & x = -1 \\ 0.5, & x = 1 \\ 0 & \text{elsewhere} \end{cases}$

c. $p_Y(y) = \begin{cases} 0.5, & y = -100 \\ 0.5, & y = 100 \\ 0 & \text{elsewhere} \end{cases}$

A11. With simple calculations, one can find that:

$$E[W] = E[X] = E[Y] = 0$$

Variance of A Discrete Random Variable

- For a discrete random variable X with PMF $p(x)$, the **variance** is defined as:

$$Var[X] = E\left[(X - E[X])^2\right] = \sum_{x:p(x)>0} (x - E[X])^2 p(x)$$

- The variance measures the possible variation (spread) of X.
- It calculates how far apart X would be from its mean, on the average.

Variance of A Discrete Random Variable (cont'd)

- An equivalent and more convenient formula:

$$Var[X] = E\left[X^2\right] - E[X]^2 = \left(\sum_{x:p(x)>0} x^2 p(x)\right) - \left(\sum_{x:p(x)>0} xp(x)\right)^2$$

- **Proof of Equivalency:** Denoting $\mu = E[X]$:

$$\begin{aligned} E\left[(X-\mu)^2\right] &= \sum_x (x-\mu)^2 p(x) = \sum_x \left(x^2 - 2\mu x + \mu^2\right) p(x) \\ &= \sum_x x^2 p(x) - \sum_x 2\mu x p(x) + \sum_x \mu^2 p(x) \\ &= \underbrace{\sum_x x^2 p(x)}_{E[X^2]} - 2\mu \underbrace{\sum_x xp(x)}_{E[X]=\mu} + \mu^2 \underbrace{\sum_x p(x)}_{1} \\ &= E\left[X^2\right] - \mu^2 = E\left[X^2\right] - E[X]^2 \end{aligned}$$

Variance of A Discrete Random Variable (cont'd)

- Thus, the variance of X can be calculated by using any of the following formulas:

$$Var[X] = E[(X-\mu)^2], \quad \mu = E[X]$$

$$Var[X] = E\left[X^2\right] - E[X]^2$$

both of which must yield the same result.

- Note that, variance can never be negative.

$$Var[X] \geq 0$$

Variance of A Discrete Random Variable: Example

Q12. Let X denote a random variable that takes values -1, 0, and 1 according to the following PMF:

$$p(x) = \begin{cases} 0.2, & x = -1 \\ 0.5, & x = 0 \\ 0.3, & x = 1 \\ 0, & \text{elsewhere} \end{cases}$$

a. Calculate $E[X]$.
b. Calculate $E[X^2]$.
c. Calculate the variance using $Var[X] = E[X^2] - E[X]^2$.
d. Calculate the variance using $Var[X] = E\left[(X - E[X])^2\right]$.

Variance of A Discrete Random Variable: Example (cont'd)

A12. We will use part (a) and part (b) to solve part (c).

a. $E[X] = -1 \times 0.2 + 0 \times 0.5 + 1 \times 0.3 = 0.1$

b. $E\left[X^2\right] = (-1)^2 \times 0.2 + 0^2 \times 0.5 + 1^2 \times 0.3 = 0.5$

c. $Var[X] = E\left[X^2\right] - E[X]^2 = 0.5 - (0.1)^2 = 0.5 - 0.01 = 0.49$

d. We will use part (a) to solve part (d).

$$\begin{aligned} Var[X] &= E\left[(X - E[X])^2\right] \\ &= (-1-0.1)^2 \times 0.2 + (0-0.1)^2 \times 0.5 + (1-0.1)^2 \times 0.3 \\ &= (-1.1)^2 \times 0.2 + (-0.1)^2 \times 0.5 + (0.9)^2 \times 0.3 \\ &= 1.21 \times 0.2 + 0.01 \times 0.5 + 0.81 \times 0.3 \\ &= 0.242 + 0.005 + 0.243 = 0.49 \end{aligned}$$

Variance of A Discrete Random Variable: Example II

Q13. Calculate $Var[X]$, where X represents the outcome when a fair die is rolled.

A13. We already know $E[X] = 3.5 = \frac{7}{2}$. Therefore, we first need to compute:

$$E\left[X^2\right] = 1^2 \times \frac{1}{6} + 2^2 \times \frac{1}{6} + 3^2 \times \frac{1}{6} + 4^2 \times \frac{1}{6} + 5^2 \times \frac{1}{6} + 6^2 \times \frac{1}{6} = \frac{91}{6}$$

Finally:

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{182-147}{12} = \frac{35}{12}$$

Variance of A Discrete Random Variable: Example III

Q14. Find the variance of the following three random variables W, X, and Y:

Variance of A Discrete Random Variable: Example III

Q14. Find the variance of the following three random variables W, X, and Y:

a. $p_W(w) = \begin{cases} 1, & w = 0 \\ 0, & \text{elsewhere} \end{cases}$

Variance of A Discrete Random Variable: Example III

Q14. Find the variance of the following three random variables W, X, and Y:

a. $p_W(w) = \begin{cases} 1, & w = 0 \\ 0, & \text{elsewhere} \end{cases}$

b. $p_X(x) = \begin{cases} 0.5, & x = -1 \\ 0.5, & x = 1 \\ 0 & \text{elsewhere} \end{cases}$

Variance of A Discrete Random Variable: Example III

Q14. Find the variance of the following three random variables W, X, and Y:

a. $p_W(w) = \begin{cases} 1, & w = 0 \\ 0, & \text{elsewhere} \end{cases}$

b. $p_X(x) = \begin{cases} 0.5, & x = -1 \\ 0.5, & x = 1 \\ 0 & \text{elsewhere} \end{cases}$

c. $p_Y(y) = \begin{cases} 0.5, & y = -100 \\ 0.5, & y = 100 \\ 0 & \text{elsewhere} \end{cases}$

Variance of A Discrete Random Variable: Example III

Q14. Find the variance of the following three random variables W, X, and Y:

a. $p_W(w) = \begin{cases} 1, & w = 0 \\ 0, & \text{elsewhere} \end{cases}$

b. $p_X(x) = \begin{cases} 0.5, & x = -1 \\ 0.5, & x = 1 \\ 0 & \text{elsewhere} \end{cases}$

c. $p_Y(y) = \begin{cases} 0.5, & y = -100 \\ 0.5, & y = 100 \\ 0 & \text{elsewhere} \end{cases}$

A14. With simple calculations, one can find that:

$$Var[W] = 0, \quad Var[X] = 1, \quad Var[Y] = 10000$$

Continuous Random Variables

- **Continuous RVs:** A RV is a continuous RV if it can take on a continuum (an uncountable number) of possible values.
  - X: The weight (in kg.s) of a randomly selected MU student.
    $$X \in [0, \infty)$$
  - Y: Daily profit (in dollars) of a corporation.
    $$Y \in (-\infty, \infty)$$
  - T: Water purity percentage.
    $$T \in [0, 100]$$

Describing Continuous Random Variables

- In order to describe a **continuous** random variable X mathematically, we use the following functions that belong to X:
  i. Probability **density** function of X.
  ii. Cumulative distribution function of X.
  iii. Expected value of X.
  iv. Variance of X.

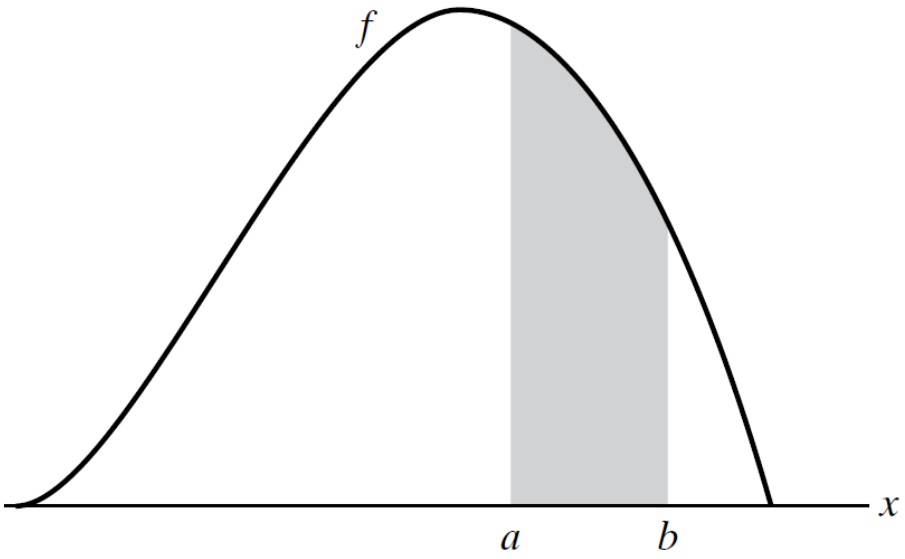
Probability Density Function (PDF)

- X is a continuous random variable if there exists a non-negative function f, defined for all $x \in \mathbb{R}$, having the property that, for any interval $[a,b]$ of real numbers:

$$P\{a \leq X \leq b\} = P\{X \in [a,b]\} = \int_a^b f(x)\,dx$$

- The function f is called the **probability density function** of the random variable X.
- Probability density function is often abbreviated as **PDF**.

Probability Density Function (PDF) (cont'd)



$P(a \leq X \leq b)$ = area of shaded region

$$P\{a \leq X \leq b\} = P\{X \in [a,b]\} = \int_a^b f(x)\,dx$$

Probability Density Function: Example

Q15. Suppose that X is a continuous random variable that denotes the amount of time (in hours) that a worker spends to setup a lathe machine. The probability density function of X is given by:

$$f(x) = \begin{cases} \frac{x}{8}, & 0 \leq x \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

What is the probability that a worker completes the setup between 2 and 3 hours?

Probability Density Function: Example

Q15. Suppose that X is a continuous random variable that denotes the amount of time (in hours) that a worker spends to setup a lathe machine. The probability density function of X is given by:

$$f(x) = \begin{cases} \frac{x}{8}, & 0 \leq x \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

What is the probability that a worker completes the setup between 2 and 3 hours?

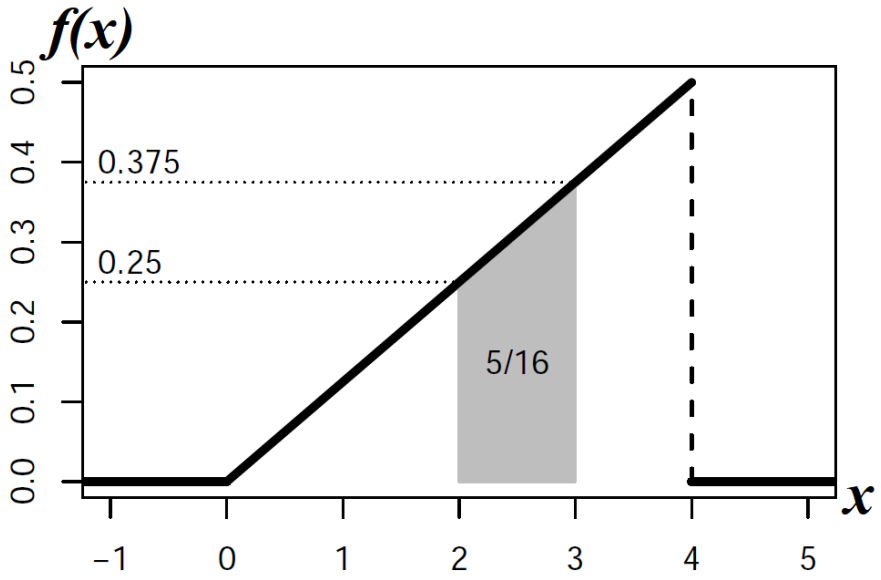
A15. The corresponding probability can be calculated by computing the definite integral of the PDF between 2 and 3 hours.

Probability Density Function: Example (cont'd)

A15. The probability is calculated as follows:

$$P\{2 \leq X \leq 3\} = \int_2^3 f(x)\,dx = \int_2^3 \frac{x}{8}dx = \left.\frac{x^2}{16}\right|_2^3 = \frac{3^2}{16} - \frac{2^2}{16} = \frac{5}{16}$$

Because the PDF is triangular, $P\{2 \leq X \leq 3\}$ can be obtained by calculating the area under the PDF between 2 and 3.



Probability Density Function: Example II

Q16. The amount of time in hours that a computer functions before breaking down is a continuous random variable with PDF given by:

$$f(x) = \begin{cases} \frac{1}{100} e^{-\frac{x}{100}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

What is the probability that:

a. a computer will function between 50 and 150 hours before breaking down?
b. it will function for less than 100 hours?

Probability Density Function: Example II (cont'd)

A16. The regarding probabilities can be computed as follows:

a.

$$\begin{aligned} P\{50 < X < 150\} &= \int_{50}^{150} \frac{1}{100} e^{-\frac{x}{100}} dx = \left[-e^{-\frac{x}{100}}\right]_{50}^{150} \\ &= \left[-e^{-\frac{3}{2}}\right] - \left[-e^{-\frac{1}{2}}\right] \\ &\approx -0.2231 + 0.6065 = 0.3834 \end{aligned}$$

b.

$$\begin{aligned} P\{X < 100\} &= \int_{0}^{100} \frac{1}{100} e^{-\frac{x}{100}} dx = \left[-e^{-\frac{x}{100}}\right]_{0}^{100} \\ &= \left[-e^{-1}\right] - [-1] \\ &\approx -0.3679 + 1 = 0.6321 \end{aligned}$$

Important Properties of A PDF

- A PDF is a **non-negative** function.
  - It can either take zero or positive values.

$$f(x) \geq 0, \quad \forall x \in \mathbb{R}$$

- The whole area under the PDF is **always** one.

$$P\{-\infty < X < \infty\} = P\{X \in (-\infty, \infty)\} = \int_{-\infty}^{\infty} f(x)\,dx = 1$$

- An important fact about CRVs is that the probability that they equal to a specific real value is always zero.

$$P\{X = a\} = \int_{a}^{a} f(x)\,dx = 0$$

An Important Fact About CRVs

- An important fact about CRVs is that the probability that they equal to a specific real value is always zero.

$$P\{X = a\} = \int_a^a f(x)\,dx = 0$$

  - We can not speak about the probability of a CRV being equal to a specific real value
  - We can only talk about the probability of a CRV being in an interval.
- An immediate result of this is:

$$P\{a \leq X \leq b\} = P\{a < X < b\} = P\{a \leq X < b\} = P\{a < X \leq b\}$$

Probability Density Function: Example III

Q17. Suppose that X is a continuous random variable that follows the probability density function:

$$f(x) = \begin{cases} c\left(2x - x^2\right), & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

a. What is the value of c?
b. Find $P\{X > 1\}$.
c. Find $P\{X = 1\}$.

A17. If $f(x)$ is really a PDF, the area under $f(x)$ in $(0, 2)$ interval should be equal to one.

Probability Density Function: Example III (cont'd)

a. We need to solve for c in $\int_0^2 c\left(2x-x^2\right)dx = 1$.

$$\int_0^2 c\left(2x-x^2\right)dx = c\left(x^2-\frac{x^3}{3}\right)\Big|_0^2 = \frac{4c}{3} = 1 \Rightarrow c = \frac{3}{4}$$

Hence, we find that the PDF of X is originally:

$$f(x) = \begin{cases} \frac{3}{4}\left(2x-x^2\right), & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Probability Density Function: Example III (cont'd)

b. We will compute $P\{X > 1\}$ by:

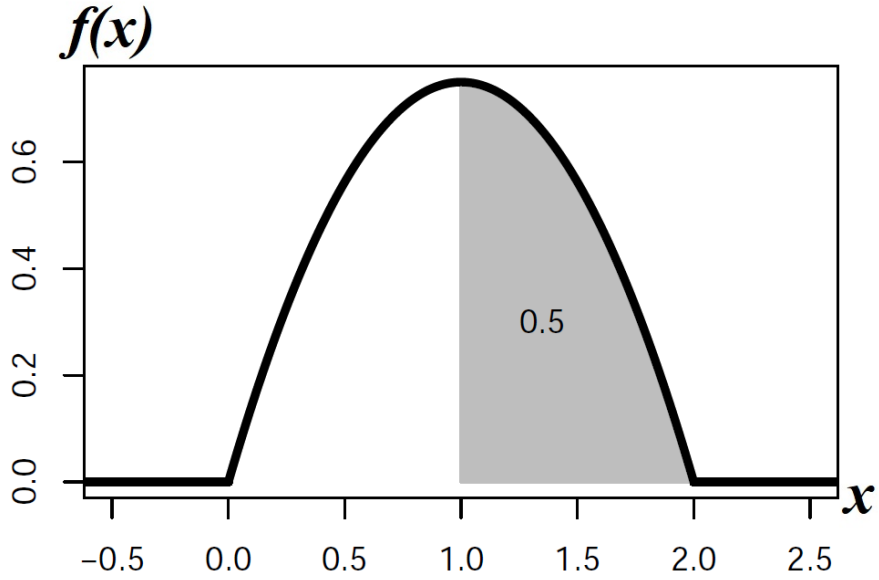
$$P\{X > 1\} = \int_{1}^{\infty} f(x)\,dx$$

However, $f(x)$ yields zero for all values above 2. Thus, we will replace the upperbound with 2.

$$\begin{aligned} P\{X > 1\} &= \int_{1}^{2} \tfrac{3}{4}\left(2x - x^2\right) dx \\ &= \tfrac{3}{4}\left(x^2 - \tfrac{x^3}{3}\right)\Big|_{1}^{2} \\ &= \tfrac{3}{4}\left[\left(4 - \tfrac{8}{3}\right) - \left(1 - \tfrac{1}{3}\right)\right] = \tfrac{1}{2} \end{aligned}$$

Probability Density Function: Example III (cont'd)

b. One can also draw the PDF and conclude that the desired probability is 0.5, since the PDF is symmetric around 1.



c. $P\{X=1\}=0$ since X is a CRV:

$$P\{X=1\}=\int_{1}^{1} f(x)\,dx=0$$

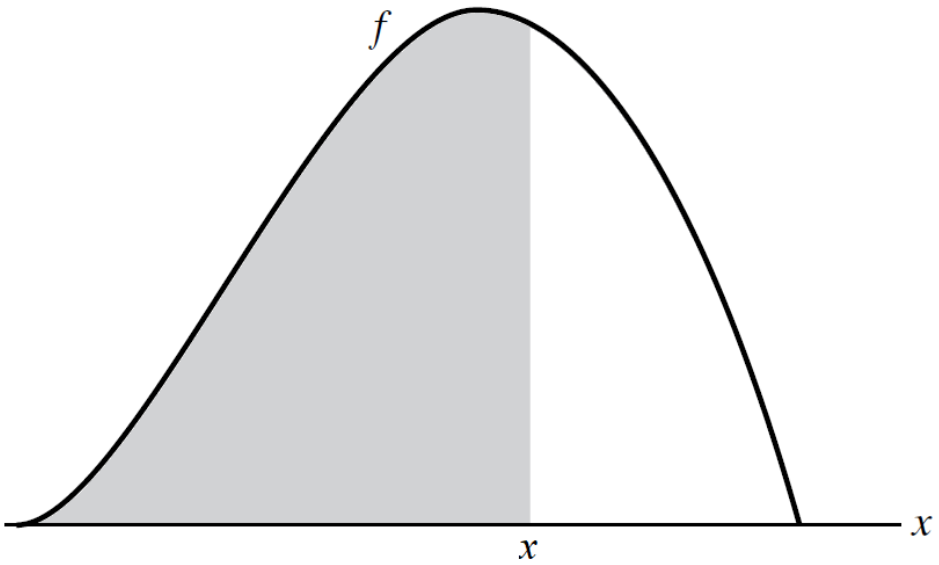
Cumulative Distribution Function (CDF)

- For a continuous random variable X, the **cumulative distribution function** is defined as:

$$F(x) = P\{X \leq x\} = P\{X < x\} = \int_{-\infty}^{x} f(t)\,dt, \quad -\infty < x < \infty$$

- It specifies, for all real values x, the probability of the random variable X to be less than or equal to a certain real value x.
- Cumulative distribution function is often abbreviated as **CDF**.

Cumulative Distribution Function (CDF) (cont'd)



$P(X \leq x)$ = area of shaded region

$$F(x) = P\{X \leq x\} = P\{X < x\} = \int_{-\infty}^{x} f(t)\,dt, \quad -\infty < x < \infty$$

Cumulative Distribution Function: Example

Q18. Suppose X is a continuous random variable with PDF:

$$f(x) = \begin{cases} \frac{1-x}{2}, & x \in [-1, 1] \\ 0, & \text{elsewhere} \end{cases}$$

a. Find the cumulative distribution function $F(x)$.
b. Use the CDF to compute $P\{X < -0.5\}$, $P\{X \leq 0\}$, and $P\{X > 0.5\}$.

Cumulative Distribution Function: Example (cont'd)

A18. Let us compute the CDF first:

a. For $x \in [-1, 1]$

$$F(x) = P\{X \leq x\} = \int_{-1}^{x} \frac{1-t}{2} dt = \left[\frac{2t-t^2}{4}\right]_{-1}^{x}$$
$$= \left[\frac{2x-x^2}{4}\right] - \left[\frac{-2-1}{4}\right] = \frac{-x^2+2x+3}{4}$$

b.

$$P\{X < -0.5\} = F(-0.5) = \frac{7}{16}$$

$$P\{X \leq 0\} = F(0) = \frac{3}{4}$$

$$P\{X > 0.5\} = 1 - P\{X \leq 0.5\} = 1 - F(0.5) = 1 - \frac{15}{16} = \frac{1}{16}$$

Important Properties of A CDF

For a continuous random variable:

- A CDF is a non-decreasing function.
  - Suppose $a, b \in \mathbb{R}$ such that $a \leq b$.
  - The PDF is a non-negative function, hence:

$$\int_{-\infty}^{a} f(t)\,dt \leq \int_{-\infty}^{b} f(t)\,dt \Rightarrow F(a) \leq F(b)$$

- When you draw a CDF, it always starts from zero and ends up at one:
  - $\lim_{x \to -\infty} F(x) = P\{X < -\infty\} = 0$
  - $\lim_{x \to \infty} F(x) = P\{X < \infty\} = 1$
- It is a **continuous** function.

Cumulative Distribution Function: Example II

Q19. Suppose X is a continuous random variable with PDF:

$$f(x) = \begin{cases} 1+x, & -1 < x \leq 0 \\ 1-x, & 0 < x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

a. Find the cumulative distribution function $F(x)$.
b. Use the CDF to compute $P\left\{-\frac{1}{2} \leq X \leq \frac{1}{2}\right\}$.
c. Draw both the PDF and the CDF.

A19. The PDF here is defined partially. Obviously, the CDF should also be computed partially.

Cumulative Distribution Function: Example II (cont'd)

a. For $-1 < x \leq 0$:

$$F(x) = \int_{-\infty}^{x} f(t)\,dt = \int_{-1}^{x} 1 + t\,dt = t + \frac{t^2}{2}\Big|_{-1}^{x} = \frac{x^2+2x+1}{2}$$

For $0 < x \leq 1$:

$$F(x) = \int_{-\infty}^{x} f(t)\,dt = \int_{-1}^{0} 1 + t\,dt + \int_{0}^{x} 1 - t\,dt = \left[t + \frac{t^2}{2}\right]_{-1}^{0} + \left[t - \frac{t^2}{2}\right]_{0}^{x} = \frac{-x^2+2x+1}{2}$$

In summary:

$$F(x) = P\{X \leq x\} = \begin{cases} 0, & -\infty < x \leq -1 \\ \frac{x^2+2x+1}{2}, & -1 < x \leq 0 \\ \frac{-x^2+2x+1}{2}, & 0 < x \leq 1 \\ 1, & 1 < x < \infty \end{cases}$$

Cumulative Distribution Function: Example II (cont'd)

b. In order to compute $P\left\{-\frac{1}{2} \leq X \leq \frac{1}{2}\right\}$, we will use the CDF:

$$P\left\{-\frac{1}{2} \leq X \leq \frac{1}{2}\right\} = P\left\{X \leq \frac{1}{2}\right\} - P\left\{X \leq -\frac{1}{2}\right\} = F\left(\frac{1}{2}\right) - F\left(-\frac{1}{2}\right)$$

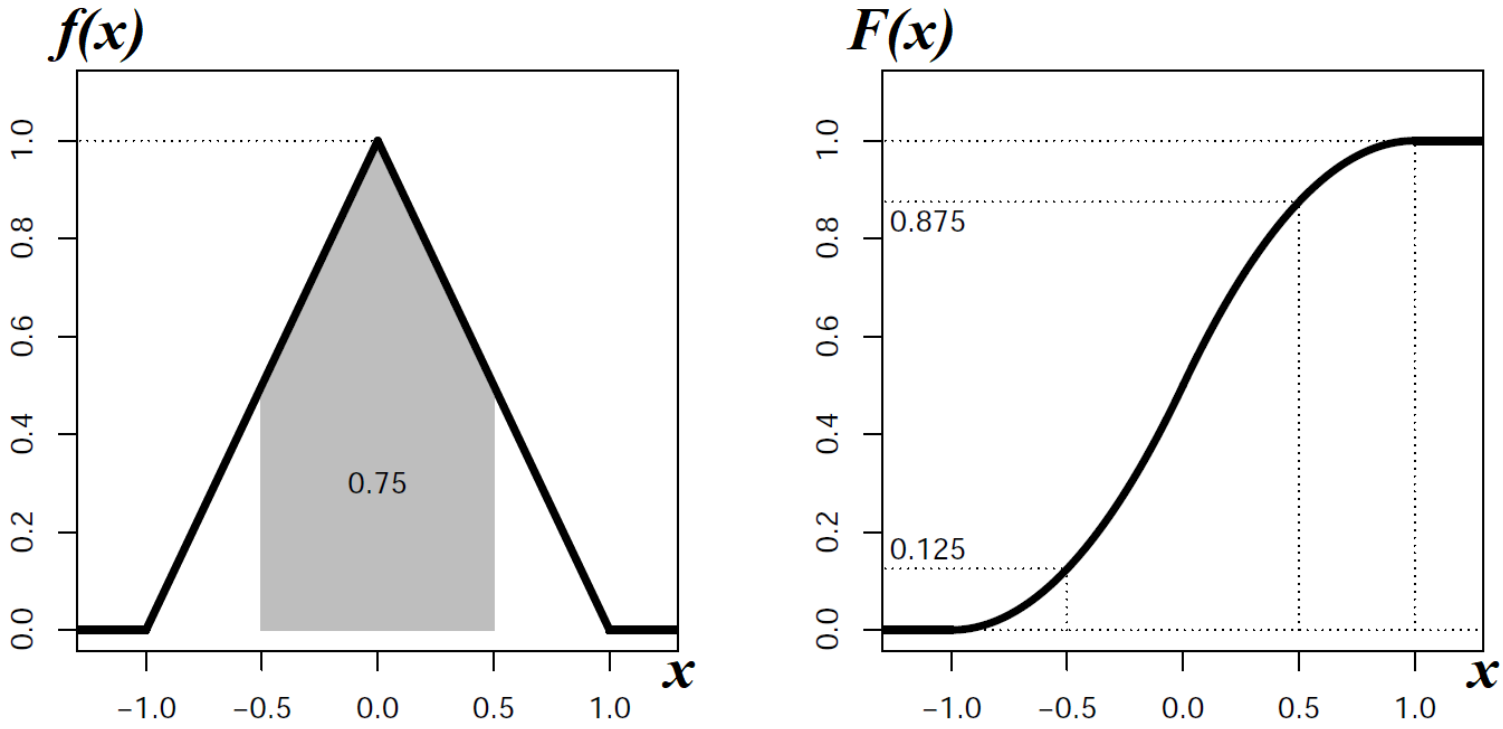
$$F\left(\frac{1}{2}\right) = \frac{-(0.5)^2+2(0.5)+1}{2} = \frac{7}{8}$$

$$F\left(-\frac{1}{2}\right) = \frac{(-0.5)^2+2(-0.5)+1}{2} = \frac{1}{8}$$

$$P\left\{-\frac{1}{2} \leq X \leq \frac{1}{2}\right\} = F\left(\frac{1}{2}\right) - F\left(-\frac{1}{2}\right) = \frac{7}{8} - \frac{1}{8} = \frac{3}{4}$$

Cumulative Distribution Function: Example II (cont'd)

c. The PDF and the CDF can be drawn as follows. Note that the probability computed in (b) is also demonstrated.



Important Properties of A CDF (cont'd)

For a continuous random variable:

- The CDF, F, is the integral of the PDF, f.

$$F(x) = P\{X \leq x\} = P\{X < x\} = \int_{-\infty}^{x} f(t)\,dt$$

- Therefore, if the CDF is differentiable, the first derivative of the CDF is equal to the PDF.

$$F'(x) = \frac{\partial F(x)}{\partial x} = f(x)$$

Cumulative Distribution Function: Example III

Q20. Suppose X is a continuous random variable with CDF:

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{x-1}{x}, & x \geq 1 \end{cases}$$

Cumulative Distribution Function: Example III

Q20. Suppose X is a continuous random variable with CDF:

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{x-1}{x}, & x \geq 1 \end{cases}$$

a. Compute $P\{2 \leq X \leq 3\}$.

Cumulative Distribution Function: Example III

Q20. Suppose X is a continuous random variable with CDF:

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{x-1}{x}, & x \geq 1 \end{cases}$$

a. Compute $P\{2 \leq X \leq 3\}$.
b. Compute the PDF $f(x)$.

Cumulative Distribution Function: Example III

Q20. Suppose X is a continuous random variable with CDF:

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{x-1}{x}, & x \geq 1 \end{cases}$$

a. Compute $P\{2 \leq X \leq 3\}$.
b. Compute the PDF $f(x)$.

Cumulative Distribution Function: Example III

Q20. Suppose X is a continuous random variable with CDF:

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{x-1}{x}, & x \geq 1 \end{cases}$$

a. Compute $P\{2 \leq X \leq 3\}$.
b. Compute the PDF $f(x)$.

A20. a. $P\{2 \leq X \leq 3\} = F(3) - F(2) = \frac{1}{6}$.

Cumulative Distribution Function: Example III

Q20. Suppose X is a continuous random variable with CDF:

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{x-1}{x}, & x \geq 1 \end{cases}$$

a. Compute $P\{2 \leq X \leq 3\}$.
b. Compute the PDF $f(x)$.

A20. a. $P\{2 \leq X \leq 3\} = F(3) - F(2) = \frac{1}{6}$.
b.

$$\frac{\partial \frac{x-1}{x}}{\partial x} = \frac{1}{x^2}$$

$$f(x) = \begin{cases} 0, & x < 1 \\ \frac{1}{x^2}, & x \geq 1 \end{cases}$$

Expected Value of A Continuous Random Variable

- For a continuous random variable X with PDF $f(x)$, the **the expected value (expectation)** is defined as:

$$E[X] = \int_{-\infty}^{\infty} x f(x)\, dx$$

- In words, the expected value of X is:
  - a weighted average of the possible values that X can take on,
  - each value being weighted by the density that X assumes it.
- Expected value is the **average (mean) value** that the random variable X can take.

Expected Value of A Continuous Random Variable: Ex.

Q21. Find $E[X]$ where the PDF of X is:

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Expected Value of A Continuous Random Variable: Ex.

Q21. Find $E[X]$ where the PDF of X is:

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

A21. Use the expectation formula for CRVs.

$$E[X] = \int_{-\infty}^{\infty} x f(x)\, dx = \int_{0}^{1} 2x^2 dx = \left[\frac{2x^3}{3}\right]_0^1 = \frac{2}{3}$$

Expected Value of A Continuous Random Variable: Ex. II

Q22. Find $E[X]$ where the PDF of X is:

$$f(x) = \begin{cases} 0, & x < 1 \\ \frac{3}{x^4}, & x \geq 1 \end{cases}$$

Expected Value of A Continuous Random Variable: Ex. II

Q22. Find $E[X]$ where the PDF of X is:

$$f(x) = \begin{cases} 0, & x < 1 \\ \frac{3}{x^4}, & x \geq 1 \end{cases}$$

A22. Use the expectation formula for CRVs.

$$E[X] = \int_{-\infty}^{\infty} x f(x)\, dx = \int_{1}^{\infty} \frac{3}{x^3} dx = \left[-\frac{3}{2x^2} \right]_1^{\infty} = 0 - \left(-\frac{3}{2} \right) = \frac{3}{2}$$

Expected Value of A Continuous Random Variable: Ex. III

Q23. Find $E[X]$ where the PDF of X is:

$$f(x) = \begin{cases} 1/4, & 0 \leq x \leq 3 \\ (5-x)/8, & 3 < x \leq 5 \\ 0, & \text{elsewhere} \end{cases}$$

Expected Value of A Continuous Random Variable: Ex. III

Q23. Find $E[X]$ where the PDF of X is:

$$f(x) = \begin{cases} 1/4, & 0 \leq x \leq 3 \\ (5-x)/8, & 3 < x \leq 5 \\ 0, & \text{elsewhere} \end{cases}$$

A23. Use the expectation formula for CRVs.

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f(x)\, dx = \int_{0}^{3} \tfrac{x}{4} dx + \int_{3}^{5} \tfrac{5x-x^2}{8} dx \\ &= \left.\tfrac{x^2}{8}\right|_0^3 + \left.\tfrac{15x^2-2x^3}{48}\right|_3^5 = \tfrac{9}{8} + \left(\tfrac{125}{48} - \tfrac{81}{48}\right) = \tfrac{49}{24} \end{aligned}$$

Variance of A Continuous Random Variable

- For a continuous random variable X with PDF $p(x)$, the **variance** is defined as:

$$Var[X] = E\left[(X - E[X])^2\right] = \int_{-\infty}^{\infty} (x - E[X])^2 f(x)\, dx$$

or with an equivalent and more convenient formula:

$$Var[X] = E\left[X^2\right] - E[X]^2 = \int_{-\infty}^{\infty} x^2 f(x)\, dx - \left(\int_{-\infty}^{\infty} x f(x)\, dx\right)^2$$

which is established in a manner similar to its counterpart in the discrete case.

Variance of A Continuous Random Variable: Example

Q24. Find $Var[X]$ where the PDF of X is:

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Variance of A Continuous Random Variable: Example

Q24. Find $Var[X]$ where the PDF of X is:

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

A24. We will use the formula $Var[X] = E\left[X^2\right] - E[X]^2$. From Question 21, we already know that $E[X] = \frac{2}{3}$.

$$E\left[X^2\right] = \int_{-\infty}^{\infty} x^2 f(x)\, dx = \int_0^1 2x^3 dx = \left[\frac{x^4}{2}\right]_0^1 = \frac{1}{2}$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}$$

Variance of A Continuous Random Variable: Example II

Q25. Find $Var[X]$ where the PDF of X is:

$$f(x) = \begin{cases} 0, & x < 1 \\ \frac{3}{x^4}, & x \geq 1 \end{cases}$$

Variance of A Continuous Random Variable: Example II

Q25. Find $Var[X]$ where the PDF of X is:

$$f(x) = \begin{cases} 0, & x < 1 \\ \frac{3}{x^4}, & x \geq 1 \end{cases}$$

A25. We will use the formula $Var[X] = E\left[X^2\right] - E[X]^2$. From Question 22, we already know that $E[X] = \frac{3}{2}$.

$$E\left[X^2\right] = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_{1}^{\infty} \frac{3}{x^2} dx = \left[-\frac{3}{x}\right]_1^{\infty} = 0 - (-3) = 3$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = 3 - \left(\frac{3}{2}\right)^2 = \frac{3}{4}$$

What is A Linear Transformation?

- For a and b being arbitrary real numbers:

$$ax + b$$

  is called a linear transformation (function) of variable x.
- **Example 1:** Fahrenheit degree is a linear transformation of the Celsius degree.

$$^{\circ}F = 1.8^{\circ}C + 32$$

- **Example 2:** The following is a linear transformation as well.

$$Y = -2X - 5, \quad (a = -2,\ b = -5)$$

Expected Value of A Linear Transformation

- **Expected Value of A Linear Transformation:** For $a, b \in \mathbb{R}$, and the random variable X:

$$E[aX+b] = aE[X]+b$$

- **Proof:** Suppose X is a continuous[2] RV with PDF f.

$$\begin{aligned} E[aX+b] &= \int_{-\infty}^{\infty} (ax+b) f(x)\, dx \\ &= \int_{-\infty}^{\infty} axf(x) + bf(x)\, dx \\ &= \int_{-\infty}^{\infty} axf(x)\, dx + \int_{-\infty}^{\infty} bf(x)\, dx \\ &= a \underbrace{\int_{-\infty}^{\infty} xf(x)\, dx}_{E[X]} + b \underbrace{\int_{-\infty}^{\infty} f(x)\, dx}_{1} \\ &= aE[X]+b \end{aligned}$$

[2] The proof for DRVs is similar.

Variance of A Linear Transformation

- **Variance of A Linear Transformation:** For $a, b \in \mathbb{R}$, and the random variable X:

$$Var[aX+b] = a^2 Var[X]$$

- **Proof:**

$$\begin{aligned} Var[aX+b] &= E\left[(aX+b-E[aX+b])^2\right] \\ &= E\left[(aX+b-aE[X]-b)^2\right] \\ &= E\left[a^2(X-E[X])^2\right] \\ &= a^2 E\left[(X-E[X])^2\right] \\ &= a^2 Var[X] \end{aligned}$$

Important Identities of Expectation and Variance: Example

Q26. Suppose, we modify a die by multiplying the numbers on each face with 2 and, then, subtracting 5. Let Y denote the outcome of this modified die.

Important Identities of Expectation and Variance: Example

Q26. Suppose, we modify a die by multiplying the numbers on each face with 2 and, then, subtracting 5. Let Y denote the outcome of this modified die.

a. Calculate $E[Y]$.

Important Identities of Expectation and Variance: Example

Q26. Suppose, we modify a die by multiplying the numbers on each face with 2 and, then, subtracting 5. Let Y denote the outcome of this modified die.

a. Calculate $E[Y]$.
b. Calculate $Var[Y]$.

Important Identities of Expectation and Variance: Example

Q26. Suppose, we modify a die by multiplying the numbers on each face with 2 and, then, subtracting 5. Let Y denote the outcome of this modified die.

a. Calculate $E[Y]$.
b. Calculate $Var[Y]$.

A26. Let X denote the outcome of a regular die. From Questions 9 and 13, we already know that $E[X] = \frac{7}{2}$ and $Var[X] = \frac{35}{12}$.

Important Identities of Expectation and Variance: Example

Q26. Suppose, we modify a die by multiplying the numbers on each face with 2 and, then, subtracting 5. Let Y denote the outcome of this modified die.

a. Calculate $E[Y]$.
b. Calculate $Var[Y]$.

A26. Let X denote the outcome of a regular die. From Questions 9 and 13, we already know that $E[X] = \frac{7}{2}$ and $Var[X] = \frac{35}{12}$.

We say $Y = 2X - 5$.

$$E[Y] = E[2X-5] = 2E[X] - 5 = 2 \times \frac{7}{2} - 5 = 2$$
$$Var[Y] = Var[2X-5] = 2^2 Var[X] = 2^2 \times \frac{35}{12} = \frac{35}{3}$$

Important Identities of Expectation and Variance: Ex. II

Q27. Suppose X is continuous random variable with the following PDF:

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Calculate $E\left[\frac{3-X}{4}\right]$ and $Var\left[\frac{3-X}{4}\right]$.

Important Identities of Expectation and Variance: Ex. II

Q27. Suppose X is continuous random variable with the following PDF:

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Calculate $E\left[\frac{3-X}{4}\right]$ and $Var\left[\frac{3-X}{4}\right]$.

A27. From Questions 21 and 24, we already know that $E[X] = \frac{2}{3}$ and $Var[X] = \frac{1}{18}$.

Important Identities of Expectation and Variance: Ex. II

Q27. Suppose X is continuous random variable with the following PDF:

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Calculate $E\left[\frac{3-X}{4}\right]$ and $Var\left[\frac{3-X}{4}\right]$.

A27. From Questions 21 and 24, we already know that $E[X] = \frac{2}{3}$ and $Var[X] = \frac{1}{18}$.

We say:

$$E\left[\frac{3-X}{4}\right] = E\left[-\frac{1}{4}X + \frac{3}{4}\right] = -\frac{1}{4}E[X] + \frac{3}{4} = -\frac{1}{4} \times \frac{2}{3} + \frac{3}{4} = \frac{7}{12}$$

$$Var\left[\frac{3-X}{4}\right] = Var\left[-\frac{1}{4}X + \frac{3}{4}\right] = \left(-\frac{1}{4}\right)^2 Var[X] = \frac{1}{16} \times \frac{1}{18} = \frac{1}{288}$$

END OF LECTURE SLIDE 04